UC SAN DIEGO / ECE 251C

9

Designing two-channel perfect-reconstruction filter banks

Product filters, polyphase components, and half-band design

How can one product filter determine all four reconstruction filters?

9.1 From verification to design

In Lecture 8, we verified that the Haar filters recover the input. We now reverse the question: how do we *choose* filters that reconstruct perfectly? Choosing a low-pass filter and a high-pass filter separately is not enough. Their analysis and synthesis responses must cooperate so that the alias terms cancel and the desired terms add to a delay.

We keep the same two-channel system. The analysis filters are h_0, h_1, and the synthesis filters are g_0, g_1. No coefficient processing is inserted between the banks.

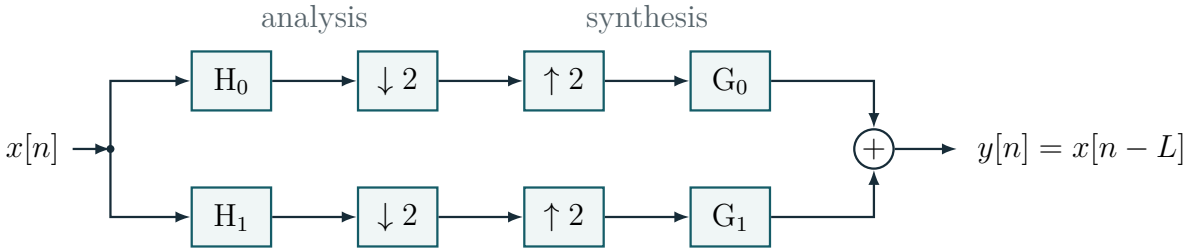


Figure 9.1: The same architecture as Lecture 8. The synthesis stage upsamples before filtering. We now design the four filters rather than verify a supplied choice.

Recall the input–output relation. With $X(z) = \sum_n x[n]z^{-n}$,

$$Y(z) = \frac{1}{2}X(z)\underbrace{\left[G_0(z)H_0(z) + G_1(z)H_1(z)\right]}_{\text{desired-signal term}} + \frac{1}{2}X(-z)\underbrace{\left[G_0(z)H_0(-z) + G_1(z)H_1(-z)\right]}_{\text{alias term}}. \tag{9.1}$$

Perfect reconstruction means $y[n] = x[n - L]$, or $Y(z) = z^{-L}X(z)$. The integer L is the overall delay; in a causal implementation we take $L \geq 0$.

9.2 Perfect reconstruction as a linear system

The two identities from Lecture 8 are

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2z^{-L}, \tag{9.2}$$
$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0. \tag{9.3}$$

The first preserves the signal up to a delay. The second removes the shifted spectrum $X(-z)$. These are the perfect-reconstruction conditions (Vetterli, 1986).

There are four unknown filter responses, but only two equations. One approach is to choose the analysis filters first, then solve for the synthesis filters:

$$\begin{bmatrix} H_0(z) & H_1(z) \\ H_0(-z) & H_1(-z) \end{bmatrix} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} = \begin{bmatrix} 2z^{-L} \\ 0 \end{bmatrix}. \tag{9.4}$$

This is a 2×2 linear system whose entries are functions of z. Its determinant is

$$\Delta(z) \coloneqq H_0(z)H_1(-z) - H_0(-z)H_1(z). \tag{9.5}$$

When Δ is not identically zero, the inverse-matrix formula gives a rational solution. Wherever the denominator is nonzero,

$$\begin{aligned} \begin{bmatrix} G_0(z) \\ G_1(z) \end{bmatrix} &= \frac{1}{\Delta(z)} \begin{bmatrix} H_1(-z) & -H_1(z) \\ -H_0(-z) & H_0(z) \end{bmatrix} \begin{bmatrix} 2z^{-L} \\ 0 \end{bmatrix} \\ &= \frac{2z^{-L}}{\Delta(z)} \begin{bmatrix} H_1(-z) \\ -H_0(-z) \end{bmatrix}. \end{aligned}$$

In particular,

$$\boxed{G_0(z) = \frac{2z^{-L}H_1(-z)}{\Delta(z)}, \qquad G_1(z) = -\frac{2z^{-L}H_0(-z)}{\Delta(z)}.} \tag{9.6}$$

The upper-right entry of the inverse has a minus sign. It multiplies zero here, but retaining it makes the full matrix identity correct.

Algebraic inversion is only the beginning. A rational expression need not describe a finite-length filter. Nor does inversion by itself guarantee a stable causal implementation. The denominator is the next design issue.

9.3 Keeping the synthesis filters finite

An FIR filter has finitely many nonzero coefficients, so its z-transform is a finite sum of integer powers of z. Multiplication by a delay z^{-r} only shifts those coefficients. Division by a nontrivial polynomial, however, can create an infinite impulse response.

If H_0, H_1 are FIR, the numerators in equation (9.6) are finite sums. The denominator Δ need not cancel. The following added example makes the distinction explicit.

A short example of the denominator problem. Take $H_0(z) = 1 - \frac{1}{2}z^{-2}$, $H_1(z) = z^{-1}$, and $L = 1$. Since $H_0(-z) = H_0(z)$ and $H_1(-z) = -z^{-1}$,

$$\Delta(z) = -2z^{-1}\big(1 - \tfrac{1}{2}z^{-2}\big),$$
$$G_0(z) = \frac{z^{-1}}{1 - \frac{1}{2}z^{-2}} = \sum_{m=0}^{\infty} 2^{-m} z^{-(2m+1)}, \qquad G_1(z) = 1.$$

The geometric expansion is valid on the unit circle. Thus $g_0[2m+1] = 2^{-m}$ for $m \geq 0$: the synthesis filter is not FIR, even though both analysis filters are. This example illustrates inversion, not a proposed low-pass/high-pass design.

A determinant that cannot create a tail. Choose

$$\boxed{\Delta(z) = 2z^{-L}.} \tag{9.7}$$

The common factor in equation (9.6) then cancels, leaving

$$\boxed{G_0(z) = H_1(-z), \qquad G_1(z) = -H_0(-z).} \tag{9.8}$$

Replacing z by $-z$ alternates the coefficient signs:

$$H(-z) = \sum_n (-1)^n h[n] z^{-n}.$$

It does not change the support. Consequently the filters in equation (9.8) are FIR whenever H_0, H_1 are FIR.

A nonzero constant times a delay also avoids a nontrivial denominator. The normalization in equation (9.7) gives the simple relations above; it is not the only convention for a PR bank.

9.4 Replace four filters by one product

Under equation (9.8), define the *product filter*

$$P_0(z) := H_0(z)G_0(z), \qquad p_0 = h_0 * g_0. \tag{9.9}$$

Since $H_1(z) = G_0(-z)$, the determinant becomes

$$\begin{aligned}\Delta(z) &= H_0(z)G_0(z) - H_0(-z)G_0(-z)\\ &= P_0(z) - P_0(-z).\end{aligned}$$

The four-filter design problem has therefore reduced to a single identity:

$$\boxed{P_0(z) - P_0(-z) = 2z^{-L}.} \tag{9.10}$$

The design procedure. First choose an FIR P_0 satisfying equation (9.10). Next factor it as $P_0 = H_0G_0$, with FIR factors. Finally set

$$H_1(z) = G_0(-z), \qquad G_1(z) = -H_0(-z). \tag{9.11}$$

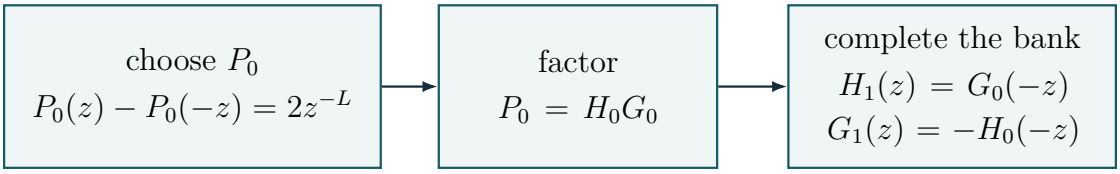


Figure 9.2: The product identity handles reconstruction. The factorization determines how that product is divided between analysis and synthesis.

Proposition 9.1. The preceding construction gives perfect reconstruction with delay L.

Proof. The alias expression cancels identically:

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = G_0(z)H_0(-z) - H_0(-z)G_0(z) = 0.$$

The desired expression is

$$G_0(z)H_0(z) + G_1(z)H_1(z) = P_0(z) - P_0(-z) = 2z^{-L}.$$

Apply equation (9.1) to obtain $Y(z) = z^{-L}X(z)$. □

Different factorizations can produce different channel filters. Perfect reconstruction is preserved, but their lengths, symmetry, gains, and frequency selectivity need not be the same. Constructing wavelet bases will require additional properties; PR alone does not supply them.

9.5 Polyphase separates even and odd taps

To design P_0, we need to understand what changes when z is replaced by $-z$. The answer is simplest when we separate the even-indexed and odd-indexed coefficients. This is the *polyphase representation*.

For a filter h, define two sequences on the coarser index grid:

$$h_{\text{even}}[m] := h[2m], \qquad h_{\text{odd}}[m] := h[2m+1].$$

The words "even" and "odd" refer to the *indices*, not to even or odd symmetry of a function. No coefficients are changed or discarded.

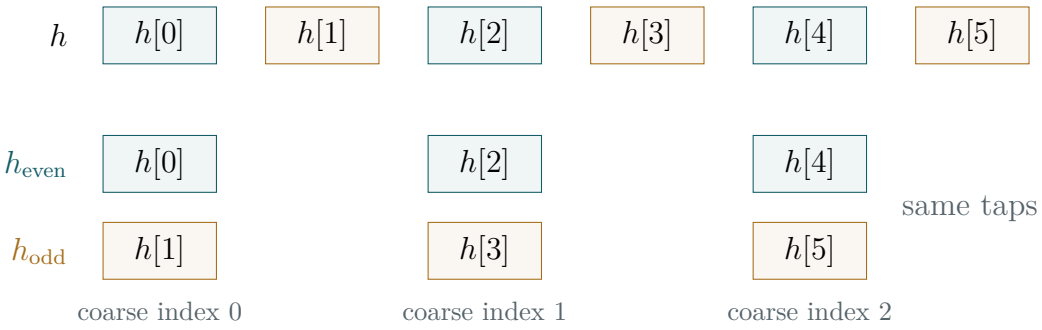


Figure 9.3: Even and odd locations become two sequences, each indexed by m. Interleaving them recovers the original filter exactly.

Separate the defining sum for $H(z)$ into its two index classes:

$$\begin{aligned} H(z) &= \sum_m h[2m] z^{-2m} + \sum_m h[2m+1] z^{-(2m+1)} \\ &= \sum_m h_{\text{even}}[m] (z^2)^{-m} + z^{-1} \sum_m h_{\text{odd}}[m] (z^2)^{-m} \\ &= H_{\text{even}}(z^2) + z^{-1} H_{\text{odd}}(z^2). \end{aligned} \tag{9.12}$$

The substitution z^2 inserts a zero between successive coefficients. The additional z^{-1} moves the odd branch onto the odd locations. In sequence notation,

$$h[n] = (\uparrow 2\, h_{\text{even}})[n] + (\uparrow 2\, h_{\text{odd}})[n-1].$$

For $n = 2m$, only the first term survives; for $n = 2m+1$, only the second survives. This checks every index in the reconstruction.

Polyphase supports efficient interpolation and antialias filtering followed by downsampling, as well as filter-bank analysis. Here its immediate purpose is algebraic: it identifies which product-filter coefficients are fixed by perfect reconstruction.

9.6 The odd component must be a delay

Apply equation (9.12) to P_0:

$$P_0(z) = P_{0,\text{even}}(z^2) + z^{-1}P_{0,\text{odd}}(z^2),$$
$$P_0(-z) = P_{0,\text{even}}(z^2) - z^{-1}P_{0,\text{odd}}(z^2).$$

Subtracting removes all even-indexed coefficients. Thus equation (9.10) is equivalent to

$$2z^{-1}P_{0,\text{odd}}(z^2) = 2z^{-L}. \tag{9.13}$$

Every power on the left has an odd exponent. Therefore L must be odd. Write

$$L = 2k + 1, \qquad k \in \mathbb{Z}.$$

After canceling $2z^{-1}$ in equation (9.13),

$$P_{0,\text{odd}}(z^2) = z^{-2k} = (z^2)^{-k}.$$

Equating coefficients, or renaming the transform variable, gives

$$\boxed{P_{0,\text{odd}}(z) = z^{-k}, \qquad p_0[2m+1] = \delta[m-k].} \tag{9.14}$$

There is exactly one nonzero odd-indexed coefficient: $p_0[L] = 1$. The even-indexed coefficients are the design variables.

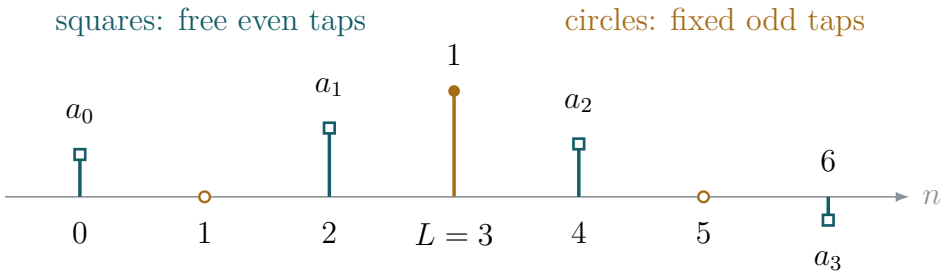


Figure 9.4: The coefficient pattern for $L = 3$. The heights of the square-marked taps are schematic design choices; the odd grid has one unit tap and zeros elsewhere.

Equivalently, we may choose any FIR even component $E(z)$ and form

$$P_0(z) = E(z^2) + z^{-L}, \qquad L \text{ odd}. \tag{9.15}$$

The product identity then holds automatically. Low-pass behavior, symmetry, and a useful factorization are additional design goals.

Why this does not contradict the previous lecture. The odd-delay conclusion belongs to the particular normalization equations (9.7) and (9.11). It is not a restriction on every PR bank. The zero-delay Haar bank from Lecture 8 used a different placement of shifts and signs.

9.7 A three-tap product and its factorization

The first example has $(p_0[0], p_0[1], p_0[2]) = (\frac{1}{2}, 1, \frac{1}{2})$, with all other taps zero. The product and its polyphase components are

$$P_0(z) = \tfrac{1}{2} + z^{-1} + \tfrac{1}{2}z^{-2},$$
$$P_{0,\text{even}}(z) = \tfrac{1}{2} + \tfrac{1}{2}z^{-1}, \qquad P_{0,\text{odd}}(z) = 1.$$

Thus $k = 0$ and $L = 1$. Direct subtraction also gives $P_0(z) - P_0(-z) = 2z^{-1}$.

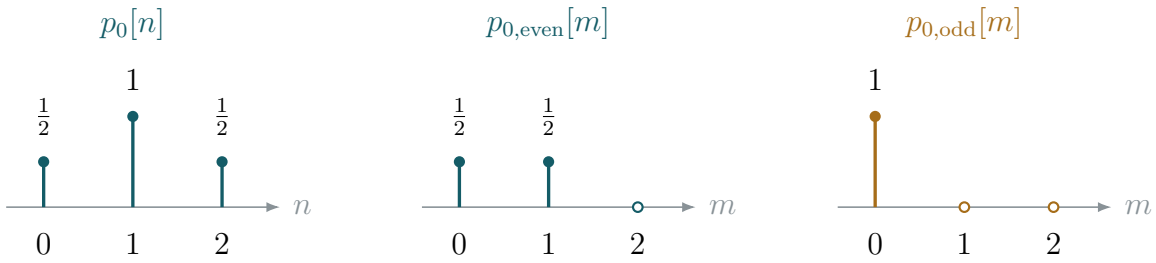


Figure 9.5: The three-tap product splits into a two-tap even component and a unit impulse on the odd grid. The latter gives $L = 2(0) + 1 = 1$.

Complete the design. As a worked continuation of this example, factor

$$P_0(z) = \tfrac{1}{2}(1 + z^{-1})^2.$$

Split the constant equally between the factors:

$$\begin{aligned} H_0(z) &= \frac{1 + z^{-1}}{\sqrt{2}}, \quad G_0(z) = \frac{1 + z^{-1}}{\sqrt{2}}, \\ H_1(z) &= \frac{1 - z^{-1}}{\sqrt{2}}, \quad G_1(z) = \frac{-1 + z^{-1}}{\sqrt{2}}. \end{aligned} \tag{9.16}$$

The second row follows from equation (9.11). These are Haar sum-and-difference filters with a causal, one-sample-delay convention. In particular, the synthesis high-pass sign is part of the construction, not a typographical choice.

Check the reconstruction on samples. If the two downsampled outputs are $c_0[m]$ and $c_1[m]$, then

$$c_0[m] = \frac{x[2m] + x[2m-1]}{\sqrt{2}}, \qquad c_1[m] = \frac{x[2m] - x[2m-1]}{\sqrt{2}}.$$

Synthesis gives

$$y[2m] = \frac{c_0[m] - c_1[m]}{\sqrt{2}} = x[2m-1], \qquad y[2m+1] = \frac{c_0[m] + c_1[m]}{\sqrt{2}} = x[2m].$$

Both index classes satisfy $y[n] = x[n-1]$.

9.8 A seven-tap product

The second example has seven taps:

$$(p_0[0], \dots, p_0[6]) = \left(-\tfrac{1}{4}, 0, \tfrac{1}{2}, 1, \tfrac{1}{2}, 0, -\tfrac{1}{4}\right).$$

Its symmetry center is $L = 3$, and its polyphase components are

$$P_{0,\text{even}}(z) = -\tfrac{1}{4} + \tfrac{1}{2}z^{-1} + \tfrac{1}{2}z^{-2} - \tfrac{1}{4}z^{-3},$$
$$P_{0,\text{odd}}(z) = z^{-1}.$$

The odd branch is again a pure delay, now with $k = 1$. Therefore

$$P_0(z) - P_0(-z) = 2z^{-3}.$$

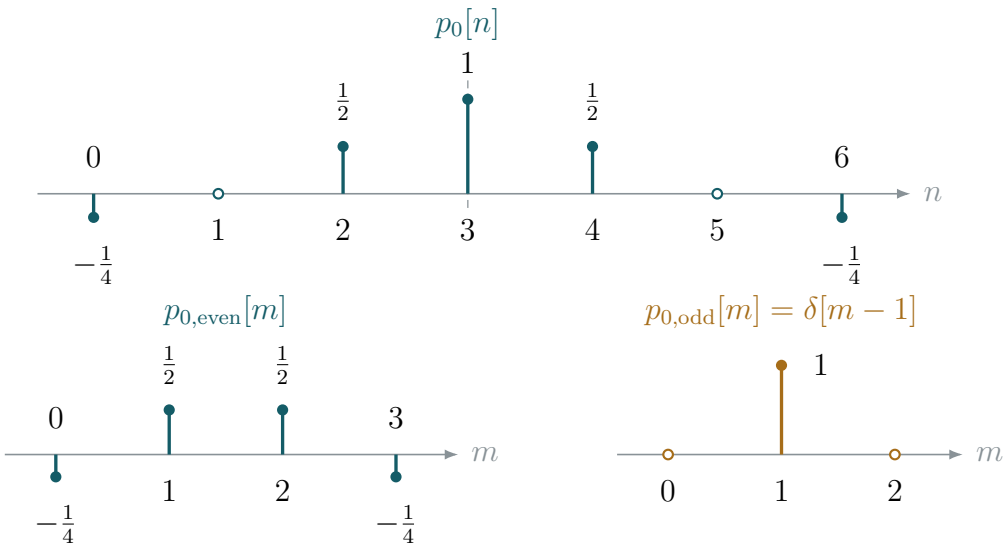


Figure 9.6: The seven-tap product and its polyphase components. The odd component has one delay, giving $L = 3$.

Again the odd grid contains one unit tap and zeros elsewhere. The even coefficients may be positive or negative. Symmetry makes this seven-tap product Type I, as in Lecture 1.

What has actually been checked? We have checked the product identity, not a low-pass specification. With the coefficients above,

$$P_0(1) = \sum_n p_0[n] = \frac{3}{2}, \qquad P_0(-1) = \sum_n (-1)^n p_0[n] = -\frac{1}{2}.$$

Their difference is two, as required. But the Nyquist response is not zero, and the DC response is not two. This is a valid *algebraic* half-band product, not an ideal low-pass response: reconstruction and frequency selectivity are distinct requirements.

9.9 Half-band filters and interpolation

The rule in equation (9.14) can be written without explicitly mentioning the odd grid. Since $L = 2k + 1$,

$$p_0[2n + L] = p_0[2(n + k) + 1] = \delta[n].$$

Definition. In our convention, a filter is *half-band centered at* L when

$$\boxed{p_0[2n + L] = \delta[n], \qquad n \in \mathbb{Z}.} \tag{9.17}$$

For our design, L is odd. The definition fixes one parity of the taps. The term "half-band" is often also used with a low-pass frequency specification; that extra goal will be imposed separately below.

Why this is an interpolation condition. Upsample a sequence x by two and filter it with p_0:

$$u[n] = \begin{cases} x[n/2], & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad r[n] = (p_0 * u)[n] = \sum_m x[m] p_0[n - 2m].$$

At the locations $n = 2\ell + L$, the output is

$$\begin{aligned} r[2\ell + L] &= \sum_m x[m] p_0[2(\ell - m) + L] \\ &= \sum_m x[m] \delta[\ell - m] = x[\ell]. \end{aligned}$$

Thus every original sample is reproduced exactly on one output phase, with delay L. The remaining output locations are filled by the other filter coefficients.

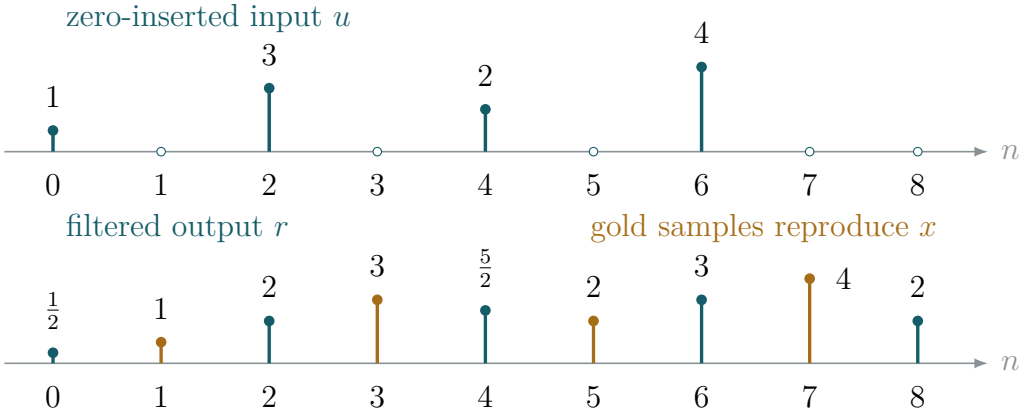


Figure 9.7: For the three-tap product, the input $(1, 3, 2, 4)$, zero elsewhere, is preserved at output indices $1, 3, 5, 7$. The intervening values are neighboring-sample averages.

Shifting the center to zero, $h[n] = p_0[n + L]$ gives the usual interpolation condition $h[2n] = \delta[n]$. Only the location of the preserved sample phase changes.

9.10 Type-I symmetry separates delay from amplitude

The product identity does not require symmetry. To obtain a linear-phase product, we now *add* the real-valued symmetry condition

$$p_0[L+r] = p_0[L-r].$$

Take a causal product supported in $\{0, \dots, 2L\}$. It has odd length and integer center L, so it is Type I. A Type-II symmetric filter has a half-integer center and does not have this centering. Here the center is the same odd integer L that appears in the product identity.

Pair equal coefficients on opposite sides of the center. For $1 \le r \le L$,

$$\begin{aligned} &p_0[L-r]\mathrm{e}^{-\mathrm{j}(L-r)\omega} + p_0[L+r]\mathrm{e}^{-\mathrm{j}(L+r)\omega} \\ &\quad = \mathrm{e}^{-\mathrm{j}L\omega} p_0[L-r]\big(\mathrm{e}^{\mathrm{j}r\omega} + \mathrm{e}^{-\mathrm{j}r\omega}\big) \\ &\quad = \mathrm{e}^{-\mathrm{j}L\omega}\, 2p_0[L-r]\cos(r\omega). \end{aligned}$$

Adding the center tap gives

$$\boxed{P_0(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}L\omega} P_{\mathrm{ampl}}(\omega),} \tag{9.18}$$

where

$$P_{\mathrm{ampl}}(\omega) = p_0[L] + 2\sum_{r=1}^{L} p_0[L-r]\cos(r\omega). \tag{9.19}$$

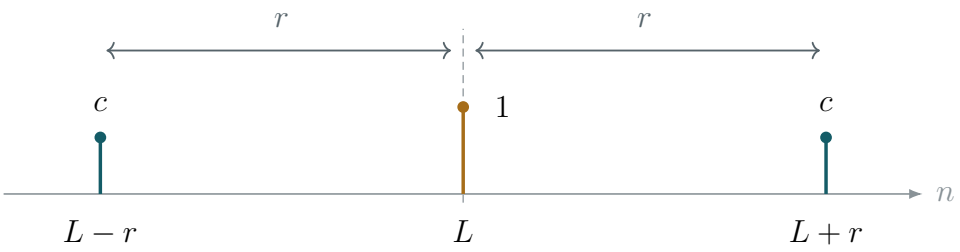


Figure 9.8: One symmetric tap pair contributes $2c\cos(r\omega)$ after the common delay $\mathrm{e}^{-\mathrm{j}L\omega}$ is removed. The center tap contributes one.

The function P_{ampl} is real, even, and 2π-periodic. It is a *signed amplitude*, not necessarily a magnitude:

$$|P_0(\mathrm{e}^{\mathrm{j}\omega})| = |P_{\mathrm{ampl}}(\omega)|.$$

This distinction matters when a stopband ripple becomes negative. Taking absolute values would destroy the signed cancellation used in the next calculation.

For the three-tap example, $P_{\mathrm{ampl}}(\omega) = 1 + \cos\omega$. For the seven-tap example, it is $1 + \cos\omega - \frac{1}{2}\cos(3\omega)$. These follow directly by pairing the displayed taps.

9.11 The two amplitudes must add to two

On the unit circle, replacing z by $-z$ shifts frequency by π:

$$-\mathrm{e}^{\mathrm{j}\omega} = \mathrm{e}^{\mathrm{j}(\omega-\pi)}.$$

Use equation (9.18) at this shifted frequency. Since L is odd,

$$\begin{aligned} P_0(-\mathrm{e}^{\mathrm{j}\omega}) &= \mathrm{e}^{-\mathrm{j}L(\omega-\pi)} P_{\mathrm{ampl}}(\omega-\pi) \\ &= \mathrm{e}^{-\mathrm{j}L\omega} \underbrace{\mathrm{e}^{\mathrm{j}L\pi}}_{-1} P_{\mathrm{ampl}}(\omega-\pi) \\ &= -\mathrm{e}^{-\mathrm{j}L\omega} P_{\mathrm{ampl}}(\omega-\pi). \end{aligned}$$

Consequently the *difference* of the two product responses becomes a *sum* of the two signed amplitudes:

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) - P_0(-\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-\mathrm{j}L\omega}\big[P_{\mathrm{ampl}}(\omega) + P_{\mathrm{ampl}}(\omega-\pi)\big].$$

Cancel the nonzero phase factor in equation (9.10) to obtain

$$\boxed{P_{\mathrm{ampl}}(\omega) + P_{\mathrm{ampl}}(\omega-\pi) = 2.} \tag{9.20}$$

Because the amplitude is even, $P_{\mathrm{ampl}}(\omega-\pi) = P_{\mathrm{ampl}}(\pi-\omega)$. On $[0,\pi]$, the two graphs are reflections about $\pi/2$.

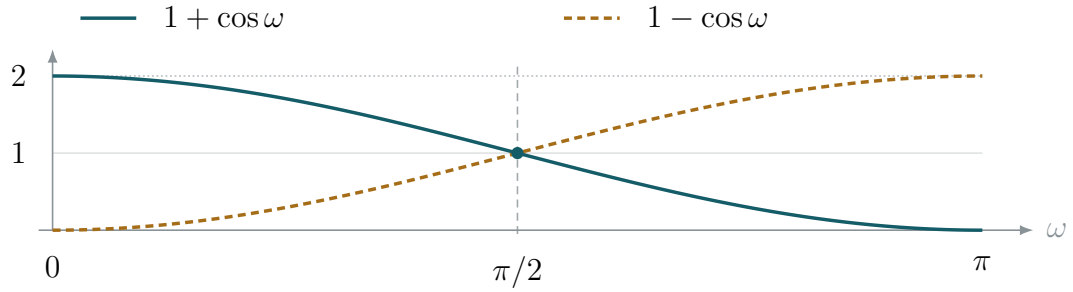


Figure 9.9: The three-tap amplitude and its shifted copy. Their sum is exactly two at every frequency, not just in the passband and stopband.

At the midpoint, the two values agree, so $P_{\mathrm{ampl}}(\pi/2) = 1$. At the endpoints, $P_{\mathrm{ampl}}(0) + P_{\mathrm{ampl}}(\pi) = 2$. For a low-pass product, we want the first endpoint value near two and the second near zero.

This is not an arbitrary functional design problem: an FIR Type-I product has the *finite cosine-polynomial* form in equation (9.19). A general complementary function need not correspond to an FIR filter.

9.12 Passbands, stopbands, and matched ripples

Let $\omega_p < \pi/2$ be a passband edge and $\omega_s > \pi/2$ a stopband edge. The desired signed amplitude is near two on $[0, \omega_p]$ and near zero on $[\omega_s, \pi]$. To pair these bands under the reflection $\omega \mapsto \pi - \omega$, choose

$$\boxed{\omega_p + \omega_s = \pi, \qquad \omega_s - \frac{\pi}{2} = \frac{\pi}{2} - \omega_p.} \tag{9.21}$$

These are symmetric specifications around half the Nyquist frequency. They do not, by themselves, guarantee perfect reconstruction: the full amplitude identity must hold at *every* frequency, including the transition band.

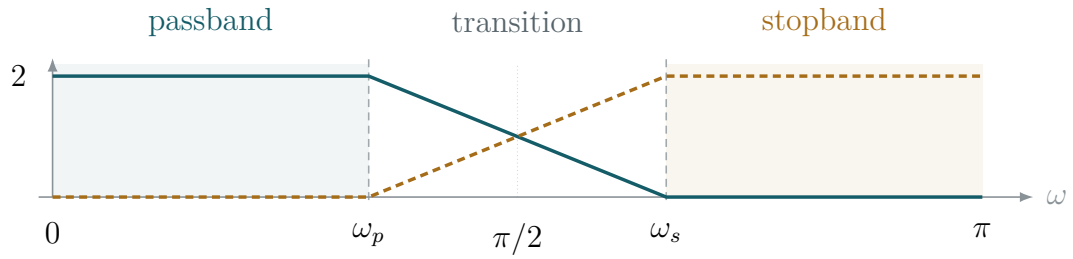


Figure 9.10: Complementary amplitude specifications. Reflection exchanges the passband and stopband; the two band edges are symmetric about $\pi/2$.

Ripples cannot be chosen independently. If a passband value is

$$P_{\text{ampl}}(\omega) = 2 + \varepsilon(\omega),$$

then equation (9.20) forces its paired stopband value to be

$$P_{\text{ampl}}(\pi - \omega) = -\varepsilon(\omega).$$

A positive passband error requires an equally large negative stopband error. The signed errors cancel; their magnitudes do not.

9.13 Build complementarity into the FIR coefficients

The amplitude identity looks like a constraint at infinitely many frequencies. The coefficient pattern turns it into a finite design problem.

For a symmetric product centered at odd L, we already know $p_0[L] = 1$. If $r > 0$ is even, then $L - r$ is an odd index different from L, so $p_0[L - r] = 0$. The cosine expansion therefore contains only odd harmonics:

$$\boxed{P_{\text{ampl}}(\omega) = 1 + \sum_{\substack{1 \le r \le L \\ r \text{ odd}}} b_r \cos(r\omega), \qquad b_r = 2p_0[L - r].} \tag{9.22}$$

Conversely, choose real coefficients b_r in this finite sum and set $p_0[L \pm r] = b_r/2$. Each odd harmonic changes sign after a π shift:

$$\cos\big(r(\omega - \pi)\big) = (-1)^r \cos(r\omega) = -\cos(r\omega).$$

Thus all the cosine terms cancel in the sum of the shifted amplitudes, leaving two. The product identity is automatic. We can select the free coefficients to improve the low-pass shape without losing that identity.

A concrete ripple illustration. The following added example is a finite polynomial, not an optimized design:

$$R(\omega) = 1 + \frac{4}{\pi} \sum_{r=0}^{4} \frac{(-1)^r}{2r+1} \cos\big((2r+1)\omega\big).$$

It has odd harmonics only, so $R(\omega) + R(\omega - \pi) = 2$ exactly.

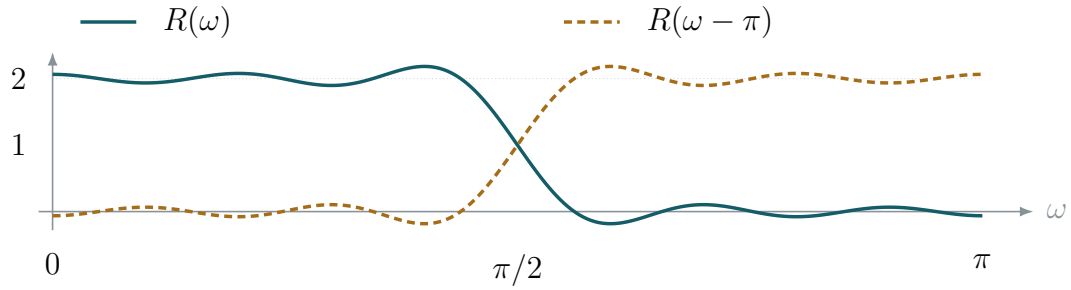


Figure 9.11: An actual finite cosine-polynomial pair. The ripples are signed and cancel exactly, including wherever one amplitude is negative.

The corresponding symmetric FIR product has center $L = 9$, center tap one, and

$$p_0[9 \pm (2r+1)] = \frac{2(-1)^r}{\pi(2r+1)}, \qquad r = 0, \ldots, 4.$$

All other taps are zero. This supplies a direct route from a finite frequency-domain choice back to an implementable product filter.

9.14 Ideal low-pass factors and the gain of two

The final example is the limiting picture in which the transition band disappears. On $[-\pi, \pi]$, set

$$C(\omega) = \mathbf{1}_{(-\pi/2, \pi/2)}(\omega),$$

and extend C periodically. Away from the band edges, $C(\omega) + C(\omega - \pi) = 1$. Values at those isolated edges do not affect the inverse DTFT.

The ideal *signed amplitude* is $P_{\text{ampl}}(\omega) = 2C(\omega)$. To retain our odd-delay convention, the full product response is

$$P_0(\mathrm{e}^{\mathrm{j}\omega}) = 2\mathrm{e}^{-\mathrm{j}L\omega}C(\omega), \qquad L \text{ odd.} \tag{9.23}$$

The rectangle in a frequency sketch depicts the amplitude, not the phase factor. The latter must remain in the filter response.

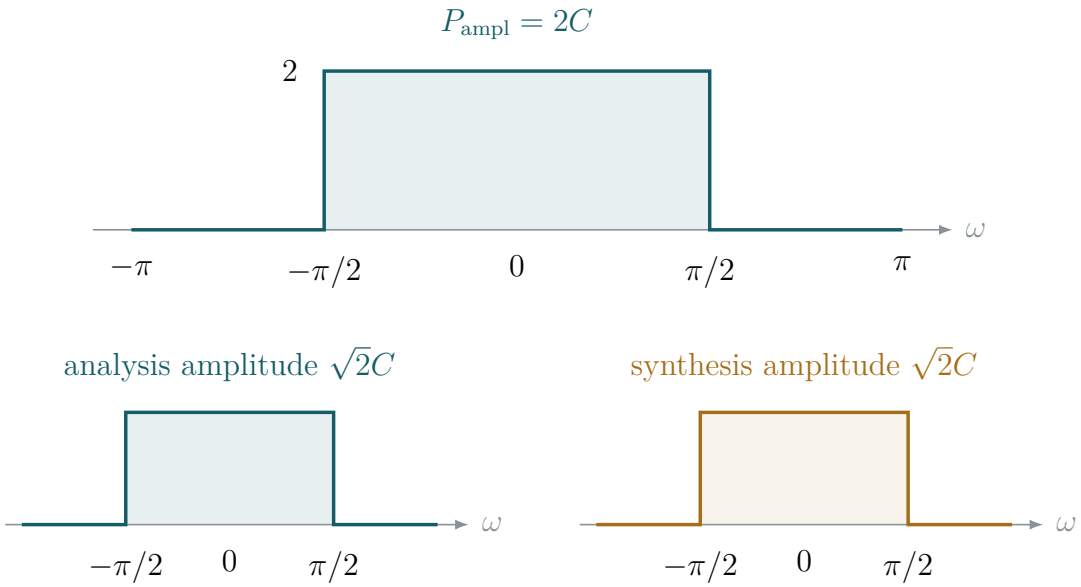


Figure 9.12: The ideal amplitude and an equal-gain factorization. Phase factors are not shown in these amplitude plots; their delays must add to L.

The gain can be split between the factors. Away from the band edges, choose any $\alpha > 0$ and integer shifts L_h, L_g with $L_h + L_g = L$:

$$H_0(\mathrm{e}^{\mathrm{j}\omega}) = \alpha\mathrm{e}^{-\mathrm{j}L_h\omega}C(\omega), \qquad G_0(\mathrm{e}^{\mathrm{j}\omega}) = \frac{2}{\alpha}\mathrm{e}^{-\mathrm{j}L_g\omega}C(\omega).$$

Since $C^2 = C$, their product is equation (9.23). The gain pairs $(2, 1)$, $(\sqrt{2}, \sqrt{2})$, and $(1, 2)$ all have product two. The complementary-band identities hold away from the band edges.

Where does the factor two come from? Recall that downsampling followed by zero insertion retains the even positions:

$$(\uparrow 2 \downarrow 2\, v)[n] = \frac{1+(-1)^n}{2}\, v[n].$$

Its z-transform is $\frac{1}{2}[V(z)+V(-z)]$, which is the source of the $1/2$ in equation (9.1). The total desired-channel response must therefore be $2z^{-L}$, not z^{-L}, to produce unit reconstruction gain.

This does *not* mean that downsampling divides every retained sample by two: $(\downarrow 2\, v)[m] = v[2m]$. The factor $1/2$ appears in the spectral decomposition of the even-index selector.

The ideal product is not FIR. Inverting equation (9.23) makes this explicit:

$$\begin{aligned} p_0[n] &= \frac{1}{2\pi}\int_{-\pi}^{\pi} P_0(\mathrm{e}^{\mathrm{j}\omega})\mathrm{e}^{\mathrm{j}\omega n}\,\mathrm{d}\omega \\ &= \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \mathrm{e}^{\mathrm{j}\omega(n-L)}\,\mathrm{d}\omega. \end{aligned}$$

For $n \neq L$, evaluate the exponential integral:

$$p_0[n] = \frac{1}{\pi}\frac{\mathrm{e}^{\mathrm{j}\pi(n-L)/2} - \mathrm{e}^{-\mathrm{j}\pi(n-L)/2}}{\mathrm{j}(n-L)} = \frac{2\sin\big(\pi(n-L)/2\big)}{\pi(n-L)}.$$

At $n = L$, the integral is one. Thus

$$\boxed{p_0[n] = \operatorname{sinc}\left(\frac{n-L}{2}\right), \qquad \operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}, \quad \operatorname{sinc}(0) = 1.} \tag{9.24}$$

It satisfies $p_0[2m+L] = \operatorname{sinc}(m) = \delta[m]$, but has infinitely many nonzero taps on the other parity. The ideal frequency picture is therefore a design target, not an FIR realization.

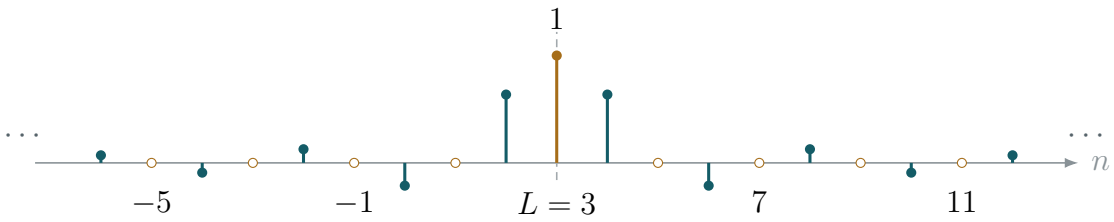


Figure 9.13: The ideal product impulse response for $L = 3$. Odd-indexed taps vanish except at the center, while the even-indexed sinc tail continues indefinitely.

Next we impose orthogonality and connect the filter design to orthogonal wavelet systems.